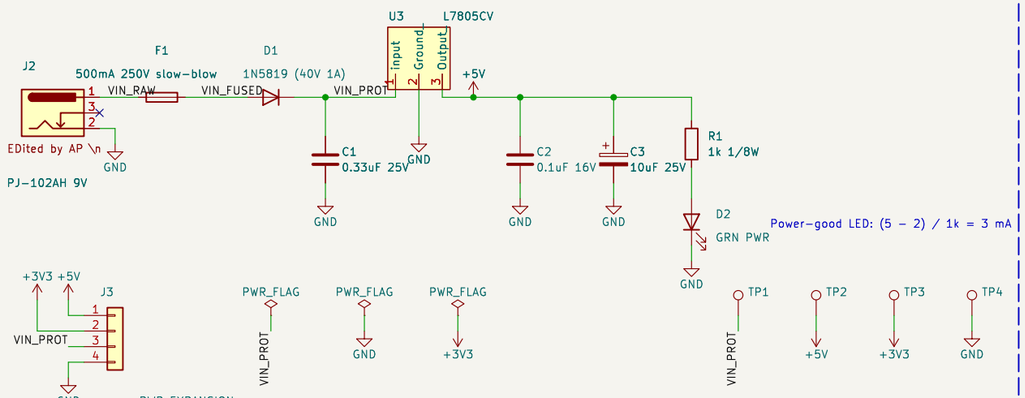
POWER INPUT 9–12 V → 5 V LINEAR REGULATOR

9 V wall adapter (Adafruit #63, 9–12 V OK) –> 0.5 A slow-blow fuse (removable, TR5 socket) –> 1N5819 reverse-polarity diode –> L7805CV (TO-220) –> +5V

Load est. < 100 mA → Pd = (12 − 5) x 0.1 = 0.7 W, OK without heatsink. LM7805 datasheet caps: 0.33 uF in, 0.1 uF out (+10 uF bulk)



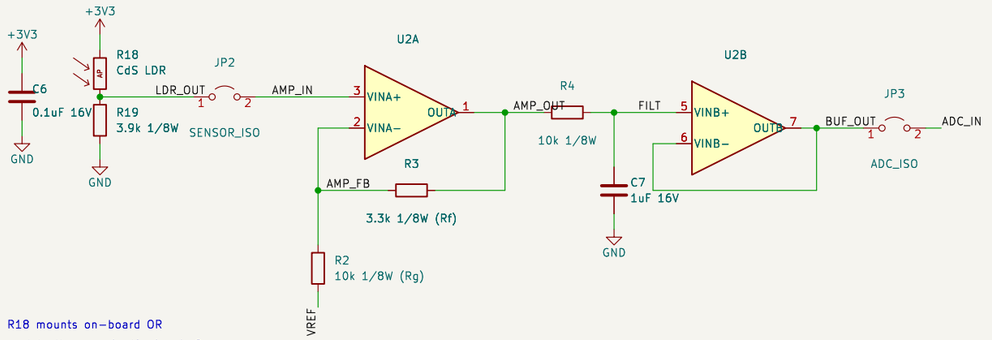
AMPLIFIED ANALOG SENSOR – PHOTORESISTOR CAP / LID SENSOR (Board B)

LDR divider on +3V3: V = 3.3 x R19/(R18 + R19). Cap closed (dark, 46k) = 0.26 V; open: low light (11k) = 0.86 V, room (1.5k) = 2.37 V.

Stage 1 U2A gain about VREF: Vout = VREF + G(Vin − VREF), G = 1 + R3/R2 = 1 + 3.3k/10k = 1.33. VREF = 0.25 V from R8/R9, buffered by U2C

Stage 2 R4/C7 passive RC low-pass: fc = 1 / (2 pi R C) = 1 / (2 pi x 10k x 1 uF) = 15.9 Hz (cap motion < 5 Hz, removes noise)

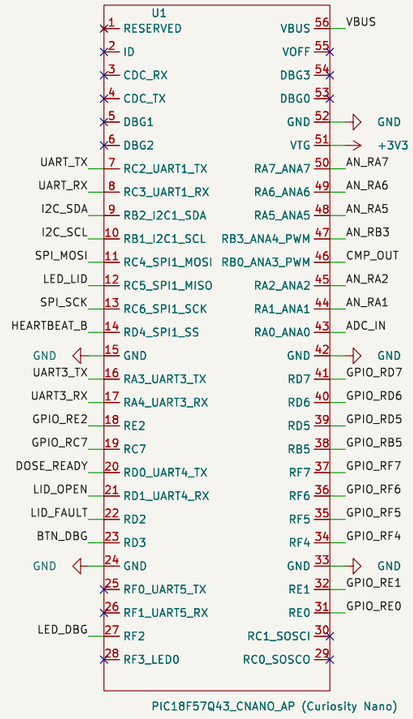
Stage 3 U2B unity-gain buffer –> low-impedance drive into ADC RA0. All analog on +3V3 so nothing can exceed the PIC pin. U2D unused.

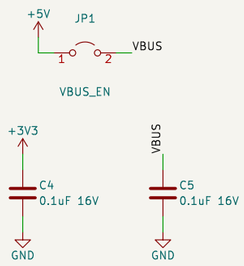


MICROCONTROLLER – PIC18F57Q43 CURIOSITY NANO (custom symbol _AP)

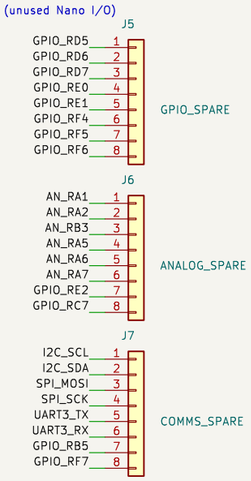
Nano runs at VTG = 3.3 V (default). +5V feeds VBUS through JP1 – REMOVE JP1 whenever the Nano USB cable is plugged in.

Hub link (J1 –> Hub J2): RD0 DOSE_READY in, RD1 LID_OPEN / RD2 LID_FAULT / RD4 HEARTBEAT_B out, UART1 RC3 RX / RC2 TX.



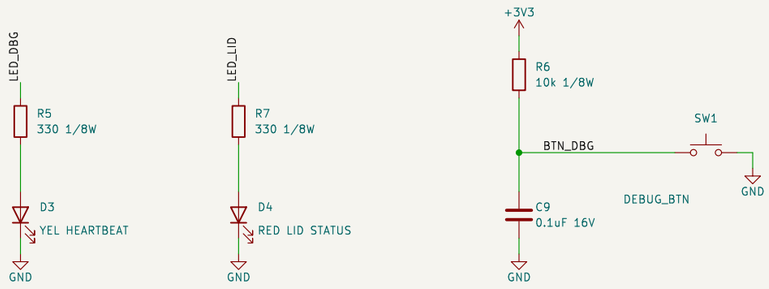


Expansion headers (unused Nano I/O)



STATUS / DEBUG LEDs + PUSH BUTTON

D3 = heartbeat on RF2 ("MCU is working"), D4 = lid status LED on RC5. I_LED = (3.3 − 2.0) / 330 = 4 mA. SW1 on RD3

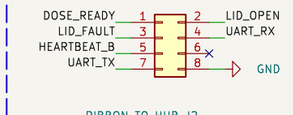


RIBBON J1 –> HUB J2

8-pin ribbon, 3.3 V logic

1–5 digital, 6–7 analog, 8 GND

Pins 4 / 7 = UART (were spare)



1 DOSE_READY hub –> B (RD0)
2 LID_OPEN B –> hub (RD1)
3 LID_FAULT B –> hub (RD2)
4 UART_RX hub –> B (RC3)
5 HEARTBEAT_B B –> hub (RD4)
6 spare (analog)
7 UART_TX B –> hub (RC2)
8 GND

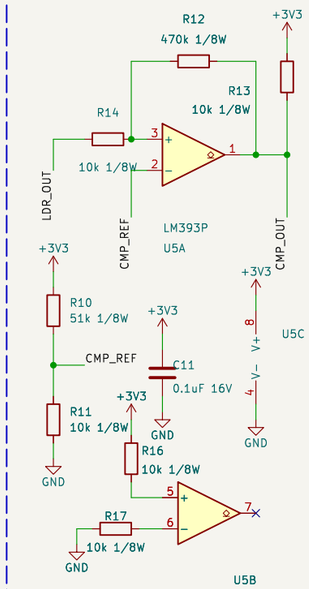
LID COMPARATOR (LM393)

Raw LDR divider vs 0.54 V threshold.

RB0 = 1 when cap OPEN (LDR_OUT > Vth)

Vth = 3.3 x 10k / 61k = 0.54 V

Hyst = 3.3 x 10k / 480k = 69 mV



PIC18F57Q43 Curiosity Nano + L7805CV + LDR/3.9k divider + MCP6004 amp/LPF/buffer + LM393 co

Arya Padiyar – Individual Subsystem Schematic – Rev 3 (photoresistor sensor)

EGR 304 Embedded Systems Design – Team 102

Sheet: /

File: BoardB_LidSensor.kicad_sch

Title: Board B – Lid Sensing Subsystem (Photoresistor)

Size: A3	Date: 2026-10-27	**Rev: 3**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1